

Mechanisms for Alternative Treatments in Parkinson's Disease: Acupuncture, Tai Chi, and Other Treatments

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Abstract At least 40 % of patients with Parkinson's disease (PD) use one or more forms of alternative therapy (AT) to complement standard treatments. This article reviews the commonest forms of AT for PD, including acupuncture, tai chi, yoga, mindfulness, massage, herbal medicine, and cannabis. We discuss the current evidence for the clinical efficacy of each AT and discuss potential mechanisms, including those suggested by animal and human studies. With a few notable exceptions, none of the treatments examined were investigated rigorously enough to draw definitive conclusions about efficacy or mechanism. Tai chi, acupuncture, *Mucuna pruriens*, cannabinoids, and music therapy have all been proposed to work through specific mechanisms, although current evidence is insufficient to support or refute these claims, with the possible exception of *Mucuna pruriens* (which contains levodopa). It is likely that most ATs predominantly treat PD patients through general mechanisms, including placebo effects, stress reduction, and improved mood and sleep, and AT may provide patients with a greater locus of control regarding their illness.

Keywords Parkinson's disease · Alternative medicine · Acupuncture · Tai chi

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Introduction

Parkinson's disease (PD) is the second commonest neurodegenerative disorder, affecting approximately 1 % of the population over the age of 60 years, and 4 % of the population over the age of 80 years in developed countries [1]. As a movement disorder, PD is characterized by four cardinal features: bradykinesia, tremor, rigidity, and postural instability. Other nonmotor symptoms include cognitive changes, depression, anxiety, apathy, sleep impairments, fatigue, autonomic dysfunction, and pain [2]. Although many motor symptoms may be related to the selective loss of dopaminergic neurons in the substantia nigra and associated changes in basal ganglia function, nondopaminergic cell loss seems to predominate and underpin nonmotor symptoms [3]. Although the cause or causes of PD are still incompletely known, significant risk factors include several known gene mutations (i.e., mutation of α -synuclein and parkin), pesticide exposure, and potentially brain injury and stress [4].

Several symptomatic therapies such as dopamine replacement and deep brain stimulation provide great benefit to PD patients. However, neither has been shown to definitely delay the progression of the disorder, nor to treat nonmotor symptoms. Physical exercise and physical therapy may benefit both motor and nonmotor symptoms of PD [5], and studies of potential neuroprotective effects in humans are under way. Owing in part to the limitations of conventional treatments, at least 40 % of PD patients use one or more forms of alternative treatment (AT) in addition to traditional treatments [6–8]. Although there is inconsistency in the definition of AT, we define it here as a treatment conceived outside or before

standard Western medical practice, including interventions developed before rigorous studies were performed to demonstrate safety or efficacy [9].

Here we review what is currently known regarding the efficacy and proposed mechanisms of the most commonly used ATs for PD. In most cases, AT is used in addition to standard treatment, and is often called complementary treatment for this reason. One area of concern is that only about half of patients using AT report use to their treating physician, and some ATs may interact with conventional approaches [6]. Although many mechanisms have been proposed for individual treatments, few empiric investigations have rigorously examined the AT mechanisms. However, results from randomized controlled clinical trials suggest that many of these treatments may improve symptoms through shared nonspecific mechanisms.

General Mechanisms of Alternative Treatments

Although anecdotal reports and uncontrolled case series of ATs frequently show positive results, randomized controlled trials frequently fail to replicate these findings, most commonly owing to a similar level of improvement in the placebo group. This suggests that many ATs may function, at least in part, through shared general mechanisms, including placebo effects, increased locus of control, and stress reduction [10].

Controlled trials reveal that up to 50 % of PD patients experienced positive placebo effects, and that placebo effectiveness was associated with treatment intensity and invasiveness [11, 12]. This response may, in part, be mediated through dopamine release in the reward system pathway related to anticipated benefits [13]. These effects appear to be modulatory rather than absolute, as placebo does not summate with dopamine-replacement therapy [13] and may additionally be effective in reducing dopaminergic-medication-induced dyskinesia [14]. Placebo mechanisms may, in part, be context- and symptom-dependent, as pain trials demonstrate distinct endorphin-mediated placebo mechanisms [15]. To fully understand the mechanisms of efficacy, AT studies should be randomized, controlled, and blinded. However, treatments that show only placebo effects should not necessarily be discounted, particularly if there are minimal risks with treatment.

Other general effects may include stress reduction and improvements in mood and sleep, all of which may impact both subjective perceptions of function and objective motor symptoms [16]. Stress has been hypothesized to impact not only acute symptoms but possibly also progression and pathogenesis [4]. Improvements can also be attributed to an increased internal locus of control, or the extent to which patients believe that their health is in their control. With this increased involvement in treatment, patients are often more optimistic, less depressed, and have a higher quality of life [17, 18].

We will now review individual ATs, including information from clinical trials and mechanistic studies. A summary of the available trials and studies is provided in Tables 1 and 2.

Acupuncture

Acupuncture has been practiced for thousands of years in traditional Asian medicine. The intention is to stimulate specific points on the body, most often with needles, in order to balance Qi, or the flow of energy throughout the body. Points are chosen on the basis of symptoms and other features (e.g., quality of pulses), and may be supplemented with electric stimulation, ointments, or cupping. Acupuncture is the third most commonly used form of AT in PD [6], and is the most studied.

In the past 15 years, several clinical trials have reported the therapeutic efficacy of acupuncture in PD. However, meta-analyses including randomized controlled trials have found that the benefits of acupuncture are generally not statistically different from those of well-designed placebo treatments, suggesting that acupuncture benefits are largely due to non-specific effects [19–21]. Larger studies including both sham acupuncture and wait-list controls will be needed to sort out potential therapeutic effects. Newer sham-acupuncture devices may allow the testing of more individualized acupuncture treatments which more accurately mimic real-world practice [55]. It is also possible that targeting of nonmotor symptoms such as fatigue and pain may show greater efficacy than targeting of motor symptoms. Notably, there were no noted adverse events or safety concerns in these studies.

Despite the inconclusive results of clinical trials, acupuncture studies in animal models report generally positive results via multiple potential mechanisms. In the unilaterally 6-hydroxydopamine (6-OHDA) lesion rat model as compared with untreated rats, acupuncture elicited significant recovery and reduced degeneration of substantia nigra dopaminergic neurons when measured by tyrosine hydroxylase staining [56]. Another study found that electroacupuncture may modulate apoptosis and neuroinflammation in the 1-methyl 4-phenyl 1,2,3,6-tetrahydropyridine (MPTP) mouse model, suggesting neuroprotective effects [57]. Several other animal studies reported similar results [58–63]. Most recently, Kim et al. [64] found the 6-OHDA lesion rat had lower levodopa needs and dyskinesia when acupuncture was added to levodopa treatment. Although these animal studies appear promising, there remain methodological concerns (e.g., controlling for handling/needling, blinding of investigators, and publication bias), particularly given discrepancies with the clinical literature.

Several studies have attempted to investigate acupuncture mechanisms on the human brain in PD by using neuroimaging. One study examined the positron emission tomography scan of PD patients, compared with a no-treatment control,

Table 1 Results of randomized controlled trials (RCTs) of alternative therapies

Treatment	RCTs $n > 50$	RCTs $n < 50$	Reported results	Adverse effects	Comments
Acupuncture	Lam et al. [19], Lee et al. [20], Lee et al. [21]	Lam et al. [19], Lee et al. [20], Lee et al. [21], Cho et al. [22]	Positive results but placebo-controlled trials generally show no effect above that of placebo	Mild (subcutaneous hematoma, dull pain, gastrointestinal symptoms)	Further rigorous investigations are needed
Tai Chi	Li et al. [23•, 24]	Cho et al. [22] Lee et al. [25], Nocera et al. [26], Amano et al. [27]	Improved motor symptoms, balance, and gait Improved motor symptoms, balance, and HRQoL	Mild (one report of itchiness) NR	Further investigation needed More recent and rigorous studies suggest benefit over other forms of exercise
Qigong	Schmitz-Hübsch et al. [28]	Burini D et al. [29]	Mixed results for motor symptoms, no other significant differences	NR	Less effective than traditional exercise in one study. Further investigation needed
Yoga	NR	Colgrove et al. [30]	Improved motor symptoms and balance	NR	Further investigations and comparisons with other exercises needed
Phytotherapy	<i>Mucuna pruriens</i> NR	Katzenschlager et al. [31]	Improvement comparable to that with LD/CD treatment, faster and longer "on" time	Mild, same as LD/CD (mild nausea, dizziness)	Further investigations on dosage and efficacy needed
	Coenzyme Q10 NR	Müller et al. [33]	Early study suggests neuroprotective effects	Mild, same as placebo	Large well-designed trial no benefit up to 1,200 mg/day
	Glutathione NR	Hauser et al. [34]	No significant differences	Mild, same as placebo	Further investigation needed
	Vitamin E Venderova et al. [35]	Carroll et al. [36]	No significant differences	NR	Not recommended
Cannabis			No significant differences in parkinsonism or dyskinesia	Mild (drowsiness, poor concentration and dry mouth)	Further investigation examining efficacy, and different active agents needed
Massage therapy	NR	Paterson et al. [37], Hernandez-Reif et al. [38], Tömhage et al. [39], Craig et al. [40]	Reduced tremor and bradykinesia, improved HRQoL and ADL	NR	Further studies needed to assess efficacy, placebo, and different forms of massage
Alexander technique	Stallibrass et al. [41]	NR	Self-reported improvements, most notably posture	NR	Further study needed including control groups
Vibration	Turbanski et al. [42], Haas et al. [43]	Kapur et al. [44], Ebersbach et al. [45], Arias et al. [46]	Improved motor symptoms	NR	Controlled trials suggest effects are nonspecific
Music therapy	NR	Pacchetti et al. [47]	Reduced bradykinesia, improved HRQoL and ADL	NR	Further investigation needed
	Dance (tango) Hackney et al. [48], Duncan et al. [49], Foster et al. [50], Hackney et al. [51]	Hackney et al. [52], McKee et al. [53], Hackney et al. [54]	Improved gait, balance HRQoL, and activity	NR	Larger blinded studies with adequate exercise controls needed

ADL activities of daily living, CD carbidopa, HRQoL health-related quality of life, LD levodopa, NR none reported

Table 2 Summary of potential mechanisms of alternative therapies

Treatment		Proposed mechanism(s)
Acupuncture		Increased glucose metabolism and basal ganglia–thalamocortical circuit activity. Reduced loss of dopaminergic neurons
Bee venom		Antiapoptotic and antiinflammatory effects
Tai chi		Specific movements to train agility and balance, greater awareness of surroundings
Qigong		Mild exercise, increased locus of control and trained internal cueing
Yoga		Specific movements to train agility and balance
Phytotherapy	<i>Mucuna pruriens</i>	High LD concentrations, MAO inhibitors, and other unknown active agents suspected of reducing substantia nigra bursts responsible for dyskinesia, antioxidant properties
	<i>Vicia faba</i>	High LD concentration
	<i>Banisteriopsis caapi</i>	MAO inhibition increases endogenous dopamine levels and reduces oxidative stress
	<i>Ginkgo biloba</i>	MAO inhibition increases endogenous dopamine levels and reduces oxidative stress
	Coenzyme Q ₁₀	Ineffective; antioxidant
	Glutathione	Ineffective; antioxidant
	Vitamin E	Ineffective, antioxidant
Cannabis		CB1 activation reduces firing bursts in the substantia nigra, antioxidant, anti-inflammatory effect of CB2 activation
Mindfulness		Increases gray matter density, increased locus of control
Massage	Therapeutic massage	Reduced stress, increased locus of control
	Reflexology	Relaxation, stress reduction
	Trager therapy	Relaxation of rigid muscles through oscillation
Alexander technique		Increased awareness, and locus of control; specific movements to train posture
Vibration		Arousal from sensory stimulation
Music therapy		External cueing, increased arousal, pleasurable stimulation
Dance		External cueing, specific movements to train agility and balance

MAO monoamine oxidase

before and after acupuncture treatment, and found a significant increase in glucose metabolism bilaterally in the frontal and occipital lobes [65]. Another small study aimed to analyze the immediate effects of acupuncture by functional magnetic resonance imaging compared with the effects of covert and overt placebos using a single acupuncture point, GB34 (the same point used in the mouse studies) [66]. Improvements in motor function were found in comparison with only the overt placebo and not with the covert placebo. Higher activity was reported in the basal ganglia–thalamocortical circuit with real acupuncture compared with the covert placebo, and comparison of covert and overt placebo treatments revealed a slightly higher activity in the anterior cingulate gyrus and superior frontal gyrus.

Cho et al. [22] performed a study in which they included a bee venom group and a no-treatment group in addition to a

standard-acupuncture group. After 8 weeks with or without treatment, they found a significant improvement on the Unified Parkinson's Disease Rating Scale as well as several nonmotor symptoms in both treatment groups. Cho et al. observed that bee venom had a greater effect on motor symptoms, whereas traditional acupuncture had the most significant effect on depression. Bee venom has been reported to inhibit MPTP-induced death of dopaminergic neurons in the mouse PD model, potentially due to reduced neuroinflammation through inhibited microglial activation [67].

Tai Chi, Yoga, and Qigong

Tai chi is a traditional Chinese martial art frequently reported to induce multiple health benefits. It includes components of aerobic exercise, stretching, balance, conscious breath control,

relaxation, and smooth gentle motions. Tai chi is one of the most widely studied and accepted forms of AT in PD, and has been reviewed previously [25]. More recently, Li et al. [23••] conducted a randomized study to compare the effects of tai chi, resistance training, and stretching on patients with PD. After 24 weeks of controlled exercise, Li et al. found that the tai chi group improved significantly on the assessment of objective motor function, had fewer falls, and retained this improvement 3 months after the intervention. On many but not all measures, tai chi was significantly better than either the balance or resistance training controls. Furthermore, the tai chi participants had a greater improvement in self-reported outcomes and were more likely to continue with some form of exercise than the resistance training or stretching groups [24]. Another small pilot study, by Nocera et al. [26], reported improvements in quality of life and mood. Tai chi has also been shown to have beneficial effects on stress in other populations [68]. Although no mechanisms have been tested for PD, one study in healthy older adults found improved neuromuscular response times when subjects were off balance [69]. However, not all trials of tai chi in PD for gait and balance reported benefits [27]. It is likely that the benefits of tai chi are a combination of physical training and conditioning with additional improvements in mood and stress. Different forms of exercise in PD are purported to induce nonoverlapping benefits [70•]; tai chi's benefits may thus, in part, be due to combining training of strength, balance, stretching, and breathing.

Yoga is another form of exercise which combines multiple physical elements with relaxation and breathing. Yoga has been shown to improve strength and flexibility in healthy adults [71], reduce fatigue in subjects with multiple sclerosis [72], improve gait and mobility in healthy geriatric subjects [73], and reduce saliva cortisol levels, a widely used stress indicator [74, 75]. Although several case studies report the benefits of yoga for PD [76–78], further investigation is scarce. Colgrove et al. [30] performed a randomized pilot study comparing patients regularly practicing a specific yoga regimen with those not using any physical intervention. They found that there was a significant improvement in motor function, posture, balance, and range of motion due to the yoga. Like tai chi, yoga should be considered a safe and potentially effective exercise regimen for PD patients; however, controlled trials examining motor and nonmotor outcomes are needed.

Qigong is a predecessor of tai chi in traditional Chinese medicine that focuses more on meditation and internal movement of energy (chi) rather than the physical exercise. The randomized controlled trials that have been performed so far on qigong have found that it provides a slight improvement in motor symptoms compared with no treatment [28], but is less effective than routine exercise [29]. Neither study found differences in nonmotor symptoms.

Herbal Treatments and Nutritional Supplementation

Vitamins and herbs are the most commonly used AT for PD [6]. The types of supplements taken differ greatly and they are often taken without physician guidance. Ayurveda, the oldest documented form of medicine in India, describes the use of *Mucuna pruriens*, or velvet beans, to treat the motor symptoms that we now distinguish as PD [79••]. Notably, *Mucuna pruriens* contains 4–6 % levodopa [80]. Katzenschlager et al. [31] examined the pharmacology of *Mucuna pruriens* compared with a standard dose of levodopa/carbidopa in a randomized, double-blind study, and found that the Ayurvedic preparation of *Mucuna pruriens* had a faster onset, and resulted in longer periods of “on time.” However, this study included only eight participants, all of whom were having difficulty with standard pharmacologic levodopa. In animal models, *Mucuna pruriens* extracts are additionally reported to have neuroprotective effects, including recovering endogenous dopamine production and reducing oxidative stress in the substantia nigra [81, 82]. *Mucuna pruriens* has also been shown to potentially result in less dyskinesia than an equivalent treatment of levodopa in animal models [83–85]. Unfortunately, as with many supplements, it is difficult to find reliable suppliers of *Mucuna pruriens*, and many purported preparations contain no detectable *Mucuna pruriens* (B.V. Manyam, personal communication). It is important, however, for patients to understand that unlike the other ATs, *Mucuna pruriens* actually contains a classical and proven treatment (levodopa).

Vicia faba, or broad bean, is another substance used in Ayurveda that also contains high concentrations of levodopa. Few studies have examined efficacy, but a small case series suggests *Vicia faba* has strong clinical dopaminergic effects, including dyskinesias [86]. *Banisteriopsis caapi* is another Ayurvedic treatment and contains harmine and harmaline, two components shown to have inhibitory effects on monoamine oxidase, but the drug has not been well studied in humans [87, 88]. The potential advantages of *Mucuna pruriens* and other Ayurvedic preparations merit further study, particularly in comparative clinical trials with standard medications.

Ginkgo biloba is a widely used herbal treatment with potential antioxidant and neuroprotective properties. Several studies have shown that *Ginkgo biloba* extract has a neuroprotective effect in MPTP animal models [89, 90]. Although there are no clinical trials in PD, studies in Alzheimer's disease suggest *Ginkgo biloba* may have modest effects on cognition when compared with placebo, but does not prevent progression to dementia in older adults [91, 92].

Coenzyme Q₁₀ is an antioxidant and electron acceptor vital to metabolic processes, and its levels are reduced in mitochondria from PD patients [93]. Although initial studies of coenzyme Q₁₀ up to a dosage of 1,200 mg/day demonstrated safety

and suggested possible neuroprotective effects, a well-designed larger study with longer longitudinal follow-up was terminated early owing to lack of efficacy [94].

Glutathione is an endogenous antioxidant, the concentration of which is also reduced in the brains of PD patients [95]. Although a small randomized controlled trial found no benefit from intravenous glutathione replacement [34], it continues to be one of the more commonly discussed and advertised ATs for PD. Similarly, although a very thorough randomized, double-blind study concluded that there was no benefit from taking vitamin E for treatment of PD [35], it continues to be the most abundantly used form of AT [6]. Although coconut oil and other nutraceuticals are often discussed by patients and on the Internet, there is currently no evidence or studies to support their use for treatment of PD.

Cannabis

Medicinal cannabis is attracting increasing interest owing to changes in legal status in many states in the USA, a growing scientific literature on the role of cannabinoid receptors in the nervous system, and publicized anecdotal reports. An anonymous survey of people with PD in the Czech Republic found that about 25 % of patients had tried using cannabis to treat their PD, and 45 % of those found it beneficial [96]. Lotan et al. [97] performed a pilot study in which patients had significant improvements in motor symptoms and reduced PD-related pain 30 min after smoking cannabis. Cannabinoids have shown mixed results in reducing the drug-induced dyskinesia [36, 98].

The cannabinoid receptor CB1 is highly expressed in the basal ganglia and has an inhibitory effect on dopaminergic pathways [99]. Thus, CB1 agonists such as Δ^9 -tetrahydrocannabinol may inhibit motor function in PD, potentially reducing tremor and dyskinesia, but also worsening other motor symptoms. One study found that Δ^9 -tetrahydrocannabinol and cannabidiol significantly reduced the depletion of dopaminergic neurons in the 6-OHDA mouse PD model [100]. It is postulated that CB1 modulates synaptic plasticity in this region, but neuroprotective effects may also be due to antioxidant properties of cannabinoids. Δ^9 -Tetrahydrocannabivarin, a CB1 antagonist with antioxidant properties, has been hypothesized to be of particular benefit in PD [101]. There are over 60 psychoactive chemicals in cannabis, including both CB1 agonists and CB1 antagonists. As different cannabis strains may have markedly different physiological and psychological effects, further mechanistic studies are needed to guide clinical trial development. Typically, practitioners discourage smoking of cannabis because of the long-term deleterious effects on the lung, and when cannabis is used, they should encourage use of an oral formulation.

Mindfulness

Mindfulness is a form of meditation in which participants are guided towards greater awareness and acceptance of the current moment [102]. In an observation of a group cognitive therapy class specifically using mindfulness for people with PD, Fitzpatrick et al. [103] concluded that mindfulness has a qualitatively beneficial effect on the patients' ability to cope. In a more mechanistic approach, Pickut et al. [104] found that 8 weeks of a mindfulness-based intervention causes structural changes in the brain of PD patients versus a no-treatment control. Through an analysis of magnetic resonance imaging scans, Pickut et al. reported significant increases in gray matter density in the cuneus and lingual gyrus of the left occipital lobe, the left thalamus, the left parahippocampus, and bilaterally in the temporoparietal junction. A randomized controlled trial of mindfulness for treatment of PD is currently under way [105].

Massage

Massage is a broad category that includes different forms of soft tissue manipulation, often incorporated with relaxation. Massage is one of the most used forms of ATs for PD [6]; however, it remains one of the least studied practices. A small pilot study of whole-body therapeutic massage in PD by Paterson et al. [37] found an improvement in quality of life and self-reported function. Reflexology, a form of therapeutic massage that focuses on the hands and feet, also resulted in improvements of nonmotor symptoms [106]. Trager therapy is a form of massage that involves a gentle rocking of the limbs. In a small study, Duval et al. [107] investigated the effect of Trager therapy on the limbs of PD patients. Using EMG, they found that the evoked stretch response in rigid limbs decreased significantly after the treatment. Another study compared massage therapy with muscle relaxation techniques and found that massage therapy resulted in significantly lower levels of stress hormones in the patient's urine, and also self-reported improvements in quality of life [38]. Although massage therapy has been shown to increase dopamine levels in the urine of other populations [108, 109], this study did not see a significant increase in PD patients. More recently, a small randomized controlled trial found that massage significantly reduced salivary cortisol levels in PD patients directly after the treatment, but had no long-lasting effect on diurnal cortisol levels [39]. Although these forms of massage have been shown to be safe and beneficial, larger clinical trials are needed to more fully document therapeutic efficacy.

Alexander Technique

The Alexander technique is a method of improving posture and movement through mental concentration and immediate

feedback from an expert [110]. It has been shown to help reduce chronic back pain, and enhance respiratory muscle function [111], two symptoms relevant to PD. Stallibras et al. [41] performed a study in which they compared the Alexander technique with massage, and with no treatment. Although the Alexander technique resulted in significant improvement in self-reported symptoms and depression compared with no treatment, there were no significant differences compared with massage. Unfortunately, this study was underpowered and did not include objective outcome measures of physical function.

Vibration Therapy

Jean-Martin Charcot first performed vibration therapy after he noted reports of improvement in motor symptoms of PD patients after train and carriage rides [44]. Since those initial reports, the efficacy of vibration therapy has been tested several times, and mechanisms have been proposed, including neuromuscular and postural adaptation. Several studies have found that whole-body vibration improves motor symptoms, especially postural stability, compared with no treatment [42, 43, 45]. Recently, two studies have examined the effect of vibration therapy against a placebo in which vibration is not applied [44, 46]. Although double blinding was not entirely possible, each study found improvement in subjects in both groups. It thus appears that vibration therapy may provide benefit through placebo or nonspecific arousal effects, and there were no adverse effects in any study.

Music Therapy

Music therapy includes a broad range of practices, from listening to music, to singing, voice exercises, playing instruments, rhythmic dancing, music-based movement, and free dancing. Several small studies have found a positive effect of music therapy on PD; this includes a meta-analysis of several small music-based movement trials finding consistent effects on several balance measures [47, 112]. One group has performed several studies investigating the benefits of dance therapy (the tango in particular) which have also reported positive effects on gait and balance [48, 49, 52–54, 113]. One study found that long-term dance participants were also more socially active [50].

Similarly to the placebo effect, music is known to activate the reward system pathway [114], releasing dopamine that could ameliorate motor symptoms temporarily. Rhythmic cueing is a tool used in physiotherapy that uses temporal auditory or visual stimulation to train specific activities [115]. For PD, rhythmic cueing has been shown to be effective in increasing gait speed and step length, and to reduce freezing [116]. It is likely that the improvements in gait through

movement-based music therapy (such as dance) may also benefit from this mechanism.

Conclusions

A large range of AT practices have been reported to have clinical or theoretical benefits for PD. Although the efficacy of many of these treatments is modest, or due to nonspecific effects, most of the studies have verified that the treatments are generally safe, and that there is no reason to discourage their use in addition to standard treatment. Notable exceptions include herbal therapies, some of which have drug interactions and/or the potential to induce dyskinesias, and also intravenously administered glutathione, which has no proven benefit but may incur significant expense along with risks from intravenous administration [117]. Overall, many of these therapies show therapeutic potential, and further mechanistic and clinical trials will be required to provide additional treatment options to PD patients.

Compliance with Ethics Guidelines

Conflict of Interest Bijan D. Ghaffari declares that he has no conflict of interest.

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